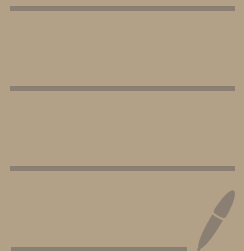


RED BLACK Tree



- (b) What is the largest possible number of internal nodes in a red-black tree with black height k ? What is the smallest possible number?

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黑高為 k 的 RB Tree 中, 最大 / 最小 內部節點數?

smallest: $2^k - 1$

largest: $2^{2k} - 1$

- (a) Show that the subtree rooted at any node x contains at least $2^{bh(x)} - 1$ internal nodes.

證以任意點為 root 的子樹, 至少包含 $2^{bh(x)} - 1$ 個內部節點

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Proof: 用樹高作歸納, 樹高 $< \infty$ 皆成立.

Basis:

樹高 $x = 0$ 時, x 是 leaf, $bh(x) = 0$, internal nodes $= 2^0 - 1 = 0 \rightarrow$ 成立

Inductive Step:

Let $bh(x) = b$

其 child 的黑高有可能是 b (紅) or $b-1$ (黑)

by IH, 每個子節點至少有 $2^{bh(x)-1} - 1$ 個 internal nodes.

$\therefore$ 包含 x 本身的 internal nodes 數為 $\geq 2 \times (2^{bh(x)-1} - 1) + 1 = 2^{bh(x)} - 1$ *

左右子樹都要算

- (a) Show that a red-black tree with n internal nodes has height $\leq 2 \lg(n+1)$.

證一擁有 n 個內部節點的 RB Tree, 其高 $h \leq 2 \lg(n+1)$

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Assume rbtree's root height $= h$, black height $= b$

by claim 1, 以任意節點為根的子樹, 至少有 $2^{bh(x)} - 1$ 個 internal nodes.

$\therefore$ 總 internal nodes $n \geq 2^{b-1}$

根據 rbtree 性質得知, red nodes $\leq \frac{h}{2}$, 表 black nodes 數至少佔一半, 即黑高 $b \geq \frac{h}{2}$

$n \geq 2^{b-1} \Rightarrow n \geq 2^{\frac{h}{2}-1}$

$= n+1 \geq 2^{\frac{h}{2}}$

$\Rightarrow \lg(n+1) \geq \lg(2^{\frac{h}{2}}) = \frac{h}{2}$

$\Rightarrow 2 \lg(n+1) \geq h$

$h \leq 2 \lg(n+1)$, 得證